

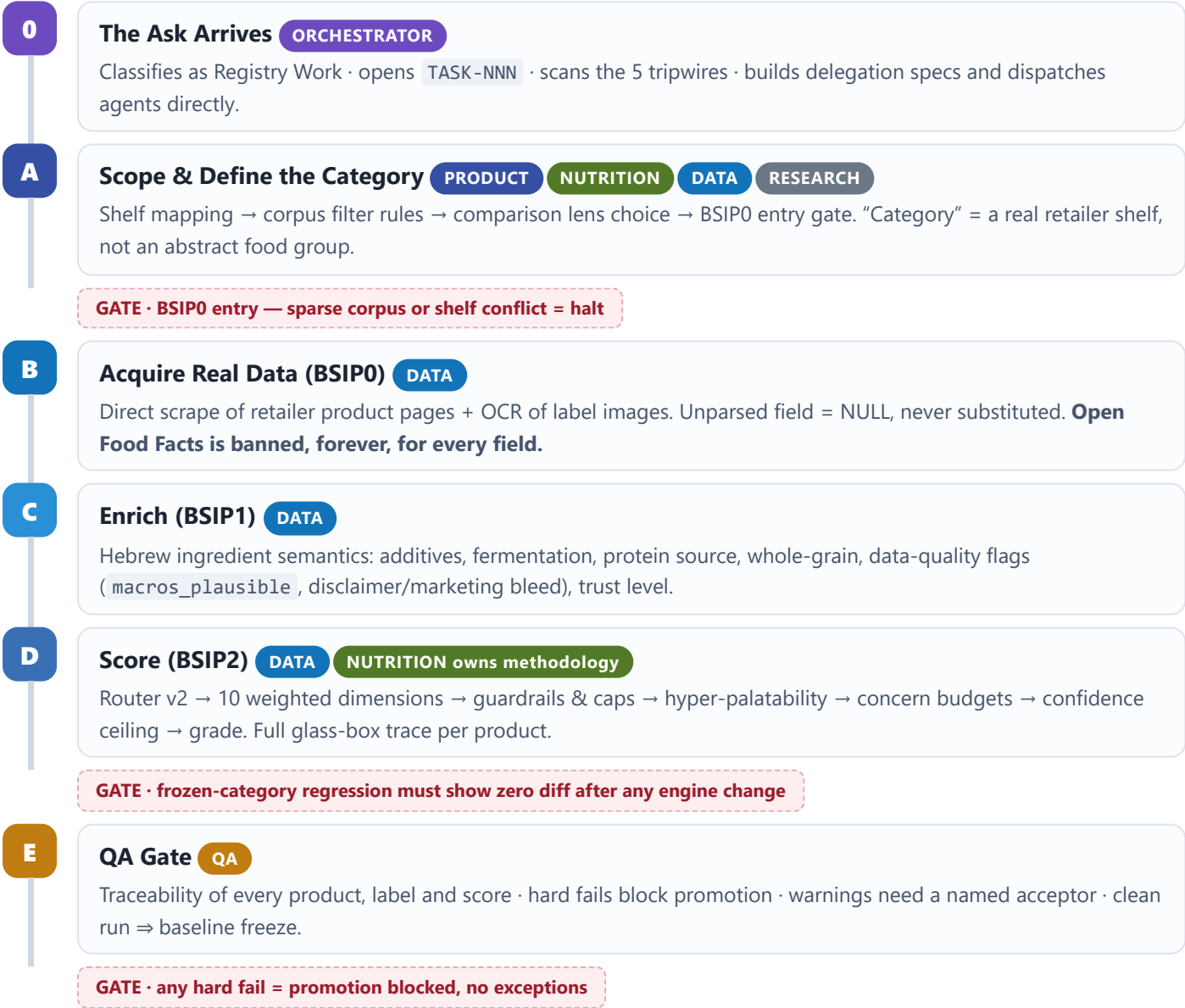
# What Happens When You Ask for a Comparison Page on “X”

The complete production chain — what the system does and what every agent does at each link, from your one-line ask to a live consumer page.

Version: v1 · Date: 2026-06-11 · Grounded in: category-factory skill · orchestration model v1 · scoring architecture (BSIP2 v0.4.1) · QA audit skill · decision authority matrix · yogurts v4 run\_006 (most recent real traversal)

- ORCHESTRATOR / CC
- PRODUCT
- NUTRITION
- RESEARCH
- DATA
- QA
- RED-TEAM
- CONTENT
- FRONTEND
- DESIGN
- MARKETING
- OWNER (YOU)

## 1 · The Chain at a Glance



**F****Red-Team Challenge** RED-TEAM

Independent adversarial audit of scores, data and claims. Findings classified CRITICAL / HIGH / MEDIUM. Criticals loop back → fix → full regeneration as a new run → re-audit.

**GATE · open CRITICAL findings block everything downstream**

**G****Consumer Copy (Hebrew)** CONTENT

Hero · 2-line row verdicts · insight lines from real trace data · mandatory category caveat box · methodology blurb. No jargon, no invented provenance.

**H****Package for the Website** DATA

Frontend JSON build → honesty blocks (implausible macros, marketing bleed) → D4 additive wiring with Hebrew explanations → byte-identical score invariant check.

**GATE · score / grade / glassBox must be byte-identical after D4 wiring**

**I****Build the Page** FRONTEND DESIGN

Route + page-data adapter + shelf filters from the 7 canonical components. The UI never sorts, rounds or interprets. `tsc` + `next build` must pass.

**J****Pre-Go-Live Gates** QA CONTENT OWNER

① QA re-verify on the live tree · ② read every consumer-facing string · ③ owner tripwire sign-off (go-live is irreversible + consumer-facing) · ④ OFF-ban sweep.

**GATE · owner sign-off — the one mandatory human checkpoint**

**K****Verify & Close** ORCHESTRATOR / CC

Every return-block claim verified against artifacts (file:line, real numbers) → `CLOSED` with evidence → unblocked tasks dispatched → page is live.

Typical calendar time for a new category: several working sessions over multiple days. Most of it goes to **B** (scrape & parse quality) and the **F→C/D loop** (red-team findings forcing rebuilds — yogurts needed 6 runs).

## 2 · Each Link in Detail

### Phase 0 — The Ask Arrives

ORCHESTRATOR (main chat)

- **Classify:** a comparison page is always *Registry Work* — multi-step, governed deliverable. Never handled inline.
- **Registry First:** check `C:\Bari\tasks\` for an existing task; otherwise open `TASK-NNN.md` (owner, status, dependencies).
- **Tripwire scan:** a new category doesn't trip a wire by itself, but two will fire later — go-live (irreversible + consumer-facing) and any touch on frozen scores.
- **Dispatch:** 5-part delegation specs (objective / boundaries / inputs / deliverable / guards) sent to owning agents — parallel where independent, background for long runs. You are never handed prompts to paste.
- **Standing rule:** after every returned deliverable, the close-readiness gate runs before anything else moves.

**Creates:** `tasks/TASK-NNN.md` · dashboard entry in `command_center.json` (auto-refresh).

### Phase A — Scoping & Category Definition

PRODUCT NUTRITION DATA RESEARCH (optional)

- **Shelf mapping:** identify the real retailer shelf slug(s) for X; verify against the shelf registry; one shelf can never feed two categories. → `shelf_map.json`
- **Corpus filter:** rules that cut the raw shelf to in-scope products, with a minimum corpus size (too sparse = halt). Boundary calls get a recorded Nutrition ruling — e.g., spread-vs-raw-chickpea is decided by tahini + sodium + energy, never by protein or the word "סלט". → `corpus_filter.json`
- **Lens & presentation:** which of the 3 comparison lenses applies; deviations from the A–E model (like frozen-veg's score-free segments) are owner-level and never a precedent. Anti-Immunity Rule: no framing may shield a product class from its real weaknesses.
- **BSIP0 entry gate:** shelf confirmed + filter non-empty + no category conflicts, or the chain halts here. → `bsip0_gate_result.json`

## Phase B — BSIP0: Data Acquisition

### DATA

- **Scrape** retailer product pages (Shufersal primary; Yohananof 4-stage: discover → approve → scrape → audit) — raw HTML/JSON + images per product.
- **OCR** physical label images through the Azure pipeline when the page lacks a parseable panel.
- **Parse** nutrition + Hebrew ingredients; the raw source is persisted for replay (lesson of the fat-overwrite bug EV-029).
- **Field honesty**: unparsed = NULL; the page will say “data could not be retrieved.” No fallback source exists, by law.

**Where it bites (yogurts):** parse *quality*, not coverage. run\_005 looked complete, but the parser was counting website disclaimer text as ingredients on 67/89 products — corrupting NOVA inference and all three 90/A grades. Only the red-team caught it. The fixes (disclaimer-strip, quality flags, plausibility gate) are now permanent infrastructure.

**Creates:** 02\_products/{cat}/observations\_bsip0/{retailer}/... + raw run JSON & log.

## Phase C — BSIP1: Semantic Enrichment

### DATA

- **Ingredient signals:** white flour vs whole grain, fermentation markers, sweeteners, protein source.
- **Additives:** E-number detection (incl. parenthesized forms) + burden count.
- **Data-quality flags:** `ingredient_text_quality` (clean / disclaimer\_stripped / marketing\_bleed), `macros_plausible` (protein = 190 g/100 g gets flagged, not scored), trust level high/medium/low.
- **Ledger:** every product is included or excluded with a named reason (yogurt run\_006: 88 in, 8 out).

**Creates:** one canonical JSON per product at 03\_operations/bsip1/run\_{cat}\_NNN/output/ + coverage report. Guarded by a 64-check pytest suite.

## Phase D — BSIP2: Scoring

DATA executes

NUTRITION owns every methodological choice

Engine v0.4.1 · per product, six stages:

| Stage                  | What happens                                                                                                                                                                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 · Features           | 50+ features from panel, ingredients, category, red labels. Missing fields recorded, <b>never imputed</b> . Router v2 assigns the structural archetype first.                                                                            |
| 2 · 10 Dimensions      | processing 15% · nutrient density 15% · calorie density 15% (category-relative: yogurt 60–250 kcal “normal”) · glycemic 12% · protein 10% · additives 10% · fat quality 8% · satiety 6% · regulatory 5% (IL red labels) · whole-food 4%. |
| 3 · Guardrails         | Trans-fat veto (floor 20) · hard caps (NOVA 4, multi red-label, sugar, sodium, additive burden — most restrictive wins) · floors for honest whole foods.                                                                                 |
| 4 · Hyper-palatability | 4 combo patterns (fat-sugar, fat-sodium, refined-carb+fat, crunch-sweet) with amplifiers and relief factors, under a family budget.                                                                                                      |
| 5 · Concern budgets    | The same root concern is never punished twice — primary signal full weight, echoes demoted.                                                                                                                                              |
| 6 · Resolution         | Caps → penalties → floors → <b>confidence ceiling</b> (null sugar –10, null sat-fat –5; low confidence can’t score high) → clamp → grade <b>before</b> rounding. S 90+ · A 80 · B 65 · C 50 · D 35 · E <35.                              |

**Standing guard:** any engine change re-runs the golden-corpus + router regressions. Frozen categories (milk 85/A top, bread, snack-bar ceiling 70/B) must show **zero diff** — otherwise the change hides behind a default-OFF env flag.

**Creates:** a glass-box `bsip2_trace.json` per product + run summary with config hash · `bsip2_readiness_checklist.json` (rule registered, observable, rollback exists).

## Phase E — QA Gate

QA

- **Traceability:** every product → corpus entry; every label → enrichment step; every score → registered rule. Any gap = hard fail.
- **Hard fails block promotion:** coverage below minimum, out-of-scope labels, unregistered rules, duplicates, version mismatch. No exceptions for small categories.
- **Warnings** (skewed distribution, low confidence, label drift) need a *named acceptor* or a fix.
- **Baseline freeze** only over a clean run; contaminated runs are invalidated with recorded reason + replacement run, and may never be referenced again.

**Creates:** `qa_gate_result.json` + audit verdict (pass / fail / invalidated) + freeze record.

## Phase F — Red-Team Challenge (mandatory, never skipped)

**RED-TEAM** — attacks only; never fixes, approves, or closes

- Hunts what the pipeline self-certified: data corruption, implausible values, misrouted products, false sweetener/fermentation detections, grade-boundary artifacts, ceiling compression, copy that overstates.
- Every finding classified **CRITICAL / HIGH / MEDIUM**; open CRITICALs block all downstream work; HIGHS must be resolved or accepted in writing.
- **The loop**: CRITICALs route to Nutrition (methodology) + Data (pipeline) → fixes → **full regeneration as a new run number** → QA re-audit → red-team re-audit. Scores are *expected* to move — that's the point.

**Creates:** 02\_products/{cat}/reports/red\_team\_{corpus}.md . QA (Phase E) checks the pipeline did what it claims; Red-Team checks whether the claims are *true*. Disjoint failure classes.

## Phase G — Consumer Copy (Hebrew)

**CONTENT**

- **Hero / prologue** — insight-first: what this shelf is and what Bari found.
- **Row verdicts** — each collapsed row is a 2-line human verdict (standing → why → catch → grade), naming calorie density + the *real fired driver* from the trace. Sodium = displayed fact only, never an unfired "driver."
- **Insight lines** — typed per spec, from actual trace data; quiet lines on unremarkable products are correct. 9 phrases banned. Jargon never leaks ("NOVA 4" → "עיבוד תעשייתי גבוה"; missing data → "ערך הסוכר לא היה זמין במקור הנתונים").
- **Category caveat** — owner-mandated "הערת קטגוריה" yellow box on every page, grounded in real engine behavior for this category.
- **Score presentation law**: numeric/grade only (72/B), no strength labels, no color-encoded judgment, 3 confidence states.

**Hard gate:** every consumer-facing string is read in full before any push — generated copy has fabricated provenance before ("official food source" shipped live, was false). Deliberate rule deviations need an Exception Registry entry *first*.

## Phase H — Frontend Packaging + D4 Additive Wiring

**DATA** (D4 Hebrew copy owned by **CONTENT**)

- **Builder script** emits per product: id, name, score, grade, insight line, confidence, image, Hebrew ingredients, nutrition, limiting factors, glass-box payload, cluster. RTL-safe; Hebrew coverage validated.
- **Honesty blocks**: `macros_plausible=false` → product blocked from the frontend; marketing-bleed ingredient text hidden rather than shown.
- **D4 wiring (mandatory)**: detect additives on every product, attach each E-number's Hebrew explanation (34-entry copy doc). Coverage <15% → halt and investigate.
- **Hard invariant**: score, grade, glassBox **byte-identical** before/after wiring — D4 annotates, never moves a score. Config hash recorded.

**Creates:** {cat}\_frontend\_vN.json → copied to bari-web/src/data/comparisons/ — **the only thing the website ever consumes from the pipeline.**

Phase I — Website Implementation

FRONTEND DESIGN reviews

- All work in `bari-web/` (Next.js) — never at the repo root.
- **Per category:** data drop + `{cat}-comparison-page-data.ts` adapter (JSON → view-model) + `{cat}-shelf-filters.ts` (typed ids + Hebrew labels) + route composed from the **7 canonical components only**.
- **Law of the view layer:** the UI never sorts, rounds, or interprets — the view-model transform is the only interpretive layer. מַעֲדוּנִים is the canonical reference; 14 Gen-0 patterns prohibited; legacy tree (milk/bread/snack) never crosses in; design tokens only, no improvised values.
- **Exit:** `tsc` green · `next build` green · RTL verified · mobile metric: shelf understood in 15–20 seconds.

Phase J — Pre-Go-Live Gates (four, in order)

QA CONTENT ORCHESTRATOR OWNER

| # | Gate                     | Holder                 | Checks                                                                                                                                                         |
|---|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | QA propagation re-verify | QA                     | Live tree == frozen run: built-site scores match BSIP2 traces, no stale JSON, routes resolve, baseline frozen on final artifacts.                              |
| 2 | Read-every-string        | Orchestrator + Content | Every consumer-facing string read in full; zero fabricated provenance/authority.                                                                               |
| 3 | Owner tripwire           | You                    | Go-live is irreversible + consumer-facing (wire 2 always fires). Also fires if any <i>published</i> grade changes (wire 1) — as with yogurts' two grade moves. |
| 4 | OFF-ban sweep            | QA                     | No Open Food Facts dependency anywhere — data, copy, images, fallbacks. Any hit = launch blocker.                                                              |

Only after all four: merge → deploy. At launch the *framing* freezes ("best ≠ excellent"); the *numbers* stay versioned and re-verify on every rescore.

Phase K — Verification & Close

ORCHESTRATOR / CC — domain agents propose RETURNED/BLOCKED; they never write CLOSED

- Every return-block claim verified **against the artifact itself** — file:line and real numbers, not prose ("88 products, 0 errors" → open the run summary and count).
- Hunt for unstated side-effects (did anything touch a frozen category? show the regression evidence).
- `status: CLOSED` recorded with a `close_reason` citing evidence; whatever the close unblocks is opened and dispatched; decision map reported.

**Durable end-state:** closed TASKs · frozen QA baseline · red-team report all-closed · run records with config hashes · memory entries for every new ruling future categories must respect.

### 3 · Who Does What — Agent Responsibility Matrix

| Agent        | Phases         | Does                                                                                                                              | Never does                                                                                            |
|--------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| ORCHESTRATOR | 0 · J · K      | Classifies, opens TASKs, dispatches delegation specs, runs the close gate after every return, closes with evidence.               | Executes pipeline work itself; hands you prompts to paste.                                            |
| PRODUCT      | A · J          | Category go/no-go, scope rationalization, lens choice, kills overbuild, signs cross-category rule adoption.                       | Writes copy; touches the engine.                                                                      |
| NUTRITION    | A · D · F-loop | Category interpretation, boundary rulings, owns scoring methodology and every ruling, resolves red-team methodology findings.     | Runs scrapers; closes own tasks; manufactures differentiation where clustering is the honest finding. |
| RESEARCH     | A · F-loop     | Evidence for rules (literature, USDA FDC; Tzameret directional-only), market context.                                             | Makes decisions — evidence only.                                                                      |
| DATA         | B · C · D · H  | BSIP0 scrape + OCR, BSIP1 enrichment, BSIP2 batch runs, frontend JSON, D4 wiring, regressions.                                    | Uses OFF (banned); imputes missing fields; changes methodology without Nutrition sign-off.            |
| QA           | E · J          | QA runner, traceability, hard-fail enforcement, warning adjudication, baseline freeze/invalidate, live-tree re-verify, OFF sweep. | Accepts warnings anonymously; freezes over hard fails; skips traceability “because small category.”   |
| RED-TEAM     | F              | Independent adversarial audit; CRITICAL/HIGH/MEDIUM report.                                                                       | Fixes, approves, closes — attack-only by design.                                                      |
| CONTENT      | G · H · J      | All Hebrew consumer copy: verdicts, insight lines, caveat box, methodology, D4 additive explanations.                             | Invents data; leaks jargon; deviates from editorial law without an Exception Registry entry.          |
| FRONTEND     | I              | Route, page-data adapter, shelf filters, canonical-component composition, tsc/build green.                                        | Sorts/rounds/interprets in the UI; touches pipeline code; invents components.                         |
| DESIGN       | I              | Hierarchy, spacing, interaction review vs the canonical reference; drift detection.                                               | Implements data logic.                                                                                |
| MARKETING    | post-launch    | SEO / content / growth for the live page.                                                                                         | Gates or initiates any pipeline stage.                                                                |
| OWNER (YOU)  | A* · J         | <b>Tripwires only:</b> go-live sign-off, published-grade changes, frozen invariants, new strategic framing.                       | Everything else — autonomy-default: the system decides and logs.                                      |

### 4 · Hard Laws That Bind the Entire Chain

**1 OFF ban (absolute).** No Open Food Facts data anywhere — nutrition, ingredients, names, images, fallback, validation, copy, confidence. NULL beats borrowed data. Any dependency = launch blocker.

**2 No invented data.** Product / nutrition / ingredient facts come from the direct scrape or are absent.



**3 Frozen invariants.** Milk = run\_005\_headpin (top 85/A) · bread = real\_bread\_retail\_003\_v1 · snack-bar ceiling = 70/B. Engine changes prove zero diff on frozen categories or hide behind a default-OFF flag.

**4 Gates are sequential and non-skippable.** No packaging before QA passes · no advancement with open CRITICALs · no shipped JSON without D4 wiring.

**5 Honest findings stand.** If scores genuinely cluster, that's the finding — never add signals to manufacture differentiation (butter precedent).

**6 Separation of powers.** The engine scores, content explains, the UI displays. The UI never interprets; copy never claims what the trace didn't fire; D4 never moves a score.

**7 Closing authority is singular.** Only the orchestrator/CC records CLOSED, only after artifact-level verification.

**8 The owner touches exactly two points** in a normal run: the strategic scoping call (if novel) and the go-live tripwire. Everything else is autonomous and logged.

## 5 · Worked Example — Yogurts v4, the Most Recent Full Traversal

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### ● Phases B–D · run\_yogurt\_005 built

Shufersal yogurt shelf scraped; BSIP1 + BSIP2 produced run\_005; frontend v4 built. Looked clean.

### ● Phases E + F · QA FAIL + Red-Team FAIL

Red-team found the parser counting website *disclaimer text* as ingredients on 67/89 products — corrupting NOVA inference and all three 90/A grades — plus a protein=190 corruption no gate had blocked. 12 findings total.

### ● The loop · TASK-249 (Data) + TASK-250 (Nutrition)

7 parser-level fixes (disclaimer strip, plausibility gate, E414 detection, live-cultures, misroute exclusion...) + 5 methodology rulings (null-sugar –10 confidence, null-satfat –5, grade-before-round, sweetener gap, ceiling-compression disclosure).

### ● Full regeneration · run\_yogurt\_006

88 products, 0 errors · grades A:14 B:30 C:20 D:23 E:1 · one product blocked outright for implausible macros · the false 90/S fell to 89.9/A once fake fermentation keywords were stripped · 14 products changed grade.

### ● Phase G · copy corrected in the same pass

NOVA jargon out, honest missing-data lines in; category caveat update (ceiling compression) routed to Content.

### ● Phase J · current state (2026-06-11)

Three gates open before go-live: **owner sign-off** on the two published-grade changes (tripwire) · Content caveat copy · QA baseline freeze on run\_006.

### ● Phase K · pending

TASK-249 sits at RETURNED; CC verifies the return block against run artifacts before recording CLOSED.

**The takeaway:** the page was *buildable* after run\_005 — but four independent gates refused to let a corrupted corpus reach consumers, and the loop produced a measurably honest run\_006. That is the system working as designed.

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Bari Agent OS · Comparison Page Production Map v1 · 2026-06-11 · Source:

01\_framework/operations/comparison\_page\_production\_map\_v1.md · Known weak links & improvement candidates are the agreed next document.